CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 9 : Python Program to Implement Travelling Salesman Problem

Aim

To develop a Python program that finds the shortest route which visits every city exactly once and returns to the starting city.

Algorithm

- Read the number of cities and the distance matrix.
- Generate all permutations of the cities other than the starting city (city 0).
- For each permutation, calculate the total cost: start city to first city, between consecutive cities, and last city back to the start.
- Compare each cost with the best cost found so far and keep the smaller one.
- Display the minimum distance.

Program

```
import itertools

n = int(input("Enter number of cities: "))
d = []

for i in range(n):
    d.append(list(map(int, input("Enter distances: ").split())))

p = list(range(1, n))
best = 9999

for path in itertools.permutations(p):
cost = d[0][path[0]]

    for i in range(n - 2):          cost
+= d[path[i]][path[i + 1]]      cost +=
d[path[-1]][0]

    if cost < best:          best
= cost print("Minimum
distance:", best)
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/9.py ==============
Enter number of cities: 4
Enter distances: 0 10 15 20
Enter distances: 10 0 35 25
Enter distances: 15 35 0 30
Enter distances: 20 25 30 0
Minimum distance: 80
>>>
```

RESULT

Thus, the Travelling Salesman Problem was solved successfully using Python.